

TASHTYP, Russian Federation

WMO: 299560

Lat: 52.800N

Lon: 89.904E

Elev: 450

StdP: 96.04

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.4	(c) -32.2	(d) -38.0	(e) 0.1	(f) -35.3	(g) -34.9	(h) 0.1	(i) -32.1	(j) 8.6	(k) -9.5	(l) 7.5	(m) -8.8	(n) 0.2	(o) 270	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.8	18.8	27.1	17.9	25.4	17.3	20.0	26.4	19.0	25.3	18.1	23.8	2.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	13.4	22.8	16.7	12.6	21.7	15.9	11.9	20.9	59.5	26.5	56.2	25.4	53.2	24.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.8	5.9	-38.0	32.3	3.9	1.9	-40.8	33.6	-43.1	34.7	-45.3	35.8	-48.2	37.2
			-37.8	21.9	3.4	1.0	-40.3	22.6	-42.3	23.2	-44.2	23.8	-46.7	24.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-18.1	-14.7	-5.2	4.0	10.2	16.4	18.2	15.8	9.2	2.0	-6.8	-14.2		
	DBStd	13.66	7.91	7.65	6.92	4.90	4.52	3.48	2.79	3.09	4.15	4.72	6.91	7.60		
	HDD10.0	3845	872	692	471	186	55	2	0	1	63	251	503	749		
	HDD18.3	6212	1131	926	729	431	254	75	37	89	274	507	753	1007		
	CDD10.0	736	0	0	0	5	61	194	254	182	39	2	0	0		
	CDD18.3	64	0	0	0	0	1	18	33	12	1	0	0	0		
	CDH23.3	863	0	0	0	2	68	266	326	178	23	0	0	0		
	CDH26.7	186	0	0	0	0	12	64	67	39	4	0	0	0		
(14) Wind	WSAvg	2.4	1.9	1.9	2.5	3.0	2.9	2.3	2.1	1.9	2.2	2.5	2.7	2.4		
(15) Precipitation	PrecAvg	529	14	11	14	40	66	87	85	77	51	36	28	19		
	PrecMax	794	31	23	46	110	172	223	155	129	104	90	79	36		
	PrecMin	369	4	1	2	8	23	31	42	37	18	10	7	6		
	PrecStd	87	8	6	11	28	33	42	27	29	20	20	16	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.0	13.5	22.5	28.7	31.2	30.9	30.1	26.6	18.2	8.9	1.5		
		MCWB	0.4	0.5	6.9	11.5	15.9	18.5	20.1	19.4	16.8	10.6	4.6	-0.1		
	2%	DB	-1.8	0.3	9.3	19.0	25.6	28.5	28.4	27.3	22.3	14.6	5.6	-0.9		
		MCWB	-3.2	-1.7	4.7	9.8	14.4	18.2	19.4	18.2	14.5	8.8	2.9	-2.4		
	5%	DB	-4.3	-1.7	6.1	16.0	22.9	26.5	26.8	25.1	19.4	11.8	3.4	-2.2		
		MCWB	-5.5	-3.6	2.4	8.6	13.1	17.5	18.7	17.5	12.9	7.2	1.5	-3.5		
	10%	DB	-6.7	-4.0	3.7	13.1	20.1	24.4	25.1	23.2	16.9	9.3	1.7	-3.7		
		MCWB	-7.8	-5.5	0.9	7.2	11.9	16.7	18.1	16.5	11.8	5.7	0.1	-5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.9	7.1	12.2	17.1	20.3	21.8	20.6	17.4	11.2	5.1	0.3		
		MCDB	1.6	2.7	12.9	20.5	26.3	27.3	28.7	28.2	25.0	16.8	7.8	1.4		
	2%	WB	-3.1	-1.7	4.7	10.5	15.2	19.2	20.5	19.2	15.1	9.2	3.0	-2.2		
		MCDB	-1.9	0.2	8.8	17.7	23.8	26.4	26.7	25.8	20.8	13.7	5.0	-1.1		
	5%	WB	-5.4	-3.5	2.6	9.0	13.8	18.2	19.5	18.1	13.7	7.5	1.6	-3.4		
		MCDB	-4.3	-1.8	5.7	15.5	21.1	25.0	25.2	23.8	18.5	11.5	3.3	-2.2		
	10%	WB	-7.7	-5.5	1.1	7.4	12.5	17.2	18.5	17.0	12.3	5.9	0.2	-5.0		
		MCDB	-6.6	-3.9	3.5	12.8	19.0	23.2	23.8	22.1	16.2	9.1	1.7	-3.7		
(35) Mean Daily Temperature Range	MDBR		9.6	10.8	11.4	11.8	12.7	12.6	11.1	11.7	11.5	9.6	7.5	8.5		
	5% DB	MCDBR	9.6	11.4	13.7	18.5	19.7	17.1	14.7	16.3	16.9	15.1	8.9	8.1		
		MCWBR	8.6	9.8	9.2	10.5	9.8	7.9	6.6	7.9	9.1	9.5	6.6	7.2		
	5% WB	MCDBR	9.6	11.3	12.9	17.2	16.5	14.5	12.6	14.9	15.1	14.3	8.4	8.1		
		MCWBR	8.7	9.8	8.9	10.2	8.6	7.3	6.2	7.7	8.5	9.2	6.5	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.316	0.364	0.408	0.407	0.396	0.404	0.391	0.355	0.346	0.320	0.287		
	taud		2.103	2.053	2.059	2.137	2.208	2.336	2.339	2.345	2.386	2.280	2.150	2.067		
	Ebn at Noon		679	764	804	820	842	856	841	831	821	733	617	580		
	Edh at Noon		72	104	129	136	133	119	117	110	93	84	70	63		
(44) All-Sky Solar Radiation	RadAvg		1.12	2.14	3.46	4.45	5.17	5.92	5.46	4.59	3.33	1.88	0.99	0.73		
	RadStd		0.11	0.17	0.29	0.33	0.50	0.53	0.36	0.36	0.29	0.19	0.12	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page